STE Mission Planning Toolkit
One-Click Terrain Curation
Technical User Guide (with Fuser & Config Reference)

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1. Purpose and Audience

This guide explains how the One-Click Terrain workflow converts aerial/drone imagery into a simulation-ready 3D terrain package for VBS4. It is written for new users and support staff who need to understand:
• What the user does in the Toolkit
• What the system automates behind the scenes
• How distributed PhotoMesh Fusers accelerate processing
• Where configuration lives and how to onboard new machines

2. High-Level Workflow

Workflow summary: Photos in → 3D mesh out → post-process → export → package → ready for VBS4.

- User selects the imagery folder (often a network share).
- Toolkit launches PhotoMesh Wizard with enforced settings.
- PhotoMesh runs photogrammetry; if Fusers are available, compute is distributed.
- PhotoMesh outputs a mesh artifact (e.g., 3DML).
- Toolkit hands off to Reality Mesh for final processing and export to VBS4-ready outputs.
- Toolkit routes outputs to VBS4 for import and packaging.

3. Architecture Overview

The Toolkit is the orchestrator; PhotoMesh performs photogrammetry; Fusers contribute CPU; Reality Mesh finalizes outputs for VBS4.

Logical layout:

STE Toolkit UI (OneClickPanel + SettingsPanel + FuserManagement)
→ photomesh_launcher (Wizard launch, UNC utilities)
→ Network Share (\\<host_ip>\SharedMeshDrive\WorkingFuser\...)
→ PhotoMesh Wizard + PhotoMeshFuser.exe instances

→ Reality Mesh
→ VBS4 import + packaging

4. Key Files and Where They Live

These are the primary configuration and code touchpoints referenced by the system:

File	Purpose
STE_Toolkit.py	Main application (UI, One-Click orchestration, fuser management).
photomesh_launcher.py	PhotoMesh Wizard path utilities and UNC/shared path resolution.
config.ini (next to EXE)	Primary application configuration (Wizard path, fuser settings, offline/network settings).
config/fuser_config.json	Shared fuser path reference and fuser settings store.
update_photomesh_config.py	Wizard config enforcement / seeding logic.

5. End-to-End User Procedure (Step-by-Step)

Step 1 — Launch One-Click Terrain

User action:

- Open the Toolkit and select the 'One-Click Terrain' tool.

System automation:

- Toolkit initializes the workflow session and required checks.

Step 2 — Select imagery

User action:

- Select the folder containing drone imagery (typically on the network share).

System automation:

- Toolkit captures the imagery path and initializes the pipeline.

Step 3 — Start processing

User action:

- Confirm selection; Toolkit launches PhotoMesh Wizard with enforced settings.

System automation:

- Toolkit launches PhotoMesh Wizard with site-approved settings.

Step 4 — Wait for photogrammetry

User action:

- PhotoMesh runs; Fusers distribute work when detected and configured.

System automation:

- PhotoMesh begins processing; if Fusers are enabled, workload is split across machines.

Step 5 — Automatic post-processing

User action:

- Mesh output is imported into Reality Mesh for final processing/export.

System automation:

- Toolkit launches PhotoMesh Wizard with site-approved settings.

Step 6 — Export and package

User action:

- Exported VBS4-ready outputs are routed to VBS4 for import and packaging.

System automation:

- Reality Mesh exports; Toolkit routes outputs to VBS4 for import and packaging.

6. PhotoMesh Fuser System

PhotoMesh Fusers are worker processes (PhotoMeshFuser.exe) that contribute CPU to accelerate photogrammetry. All fusers write to a shared WorkingFuser directory; the system scales fuser instances to meet a configured target.

Typical scaling rules (example policy):

- User PCs can run up to 3 local fuser instances each (LocalFuser1/2/3).
- Host PC typically runs 1 instance to avoid SMB loopback and maintain coordination.

- Target counts are enforced automatically by the Toolkit based on role (host vs fuser vs neither) and work_mode.

7. Configuration Reference

Primary configuration is stored in config.ini. The examples below are representative; your site values may differ.

7.1 config.ini — [General]

```
[General]
photomesh_wizard_exe = C:\Program Files\Skyline\PhotoMesh\Tools\PhotomeshWizard\PhotoMeshWizard.exe
fast_startup = True
force_single_use_mode = False
```

7.2 config.ini — [Fusers]

```
[Fusers]
config_path = config/fuser_config.json
local_fuser_exe = C:\Program Files\Skyline\PhotoMesh\Fuser\PhotoMeshFuser.exe
fuser_computer = True
working_folder_host = [Offline.host_ip]
desired_count = 3
host_count = 1
shared_working_unc = \\[Offline.host_ip]\SharedMeshDrive\WorkingFuser
work_mode = shared
autostart_on_launch = True
kill_on_exit = True
```

7.3 config.ini — [Offline]

```
[Offline]
enabled = True
host_ip = <HOST_IP_ADDRESS>   # Single source of truth for network paths
share_name = SharedMeshDrive
local_data_root = E:\SharedMeshDrive
working_fuser_subdir = WorkingFuser
use_ip_unc = True
```

7.4 fuser_config.json (shared path)

```
{
  "shared_path": "\\\\<HOST_IP>\\SharedMeshDrive\\WorkingFuser",
  "fusers": {}
}
```

8. Onboarding Scenarios

8.1 Add a New User PC (Fuser Computer)

- Install PhotoMesh Fuser (PhotoMeshFuser.exe).
- Set [Fusers] fuser_computer = True and desired_count (1–3).
- Set [Fusers] work_mode = shared for network operation.
- Set [Offline] host_ip to the Host PC’s IP address.
- Verify the UNC path: \\<HOST_IP>\SharedMeshDrive\WorkingFuser is reachable.

8.2 Set Up a New Host PC

- Create local directory structure (example): E:\SharedMeshDrive\WorkingFuser\
- Set [Offline] host_ip to THIS PC’s IP address.
- Create SMB share SharedMeshDrive pointing to the local_data_root.
- Enable Windows File and Printer Sharing firewall rules as required.
- Host should use local paths (not UNC loopback) when resolving WorkingFuser.

9. Network Behavior Notes

Important: Windows typically cannot access its own SMB share via UNC loopback by default. The Toolkit should convert UNC to a local path when it detects it is running on the Host PC.

10. Troubleshooting and Diagnostics

Common issues and recommended actions:

- **Fusers not starting:** Verify PhotoMeshFuser.exe path exists; confirm [Fusers] fuser_computer = True; review logs/ste_toolkit.log.
- **UNC path not accessible:** Confirm [Offline] host_ip; manually test dir \\<HOST_IP>\SharedMeshDrive\WorkingFuser; verify SMB share exists (net share) and firewall rules.
- **Host loopback errors:** Confirm host is using local E:\SharedMeshDrive\WorkingFuser rather than UNC.
- **Fusers killed on exit:** Set [Fusers] kill_on_exit = False if you want fusers to persist after Toolkit closes.

10.1 Log Locations

- logs/ste_toolkit.log — main application activity
- crash_logs/ — unhandled exception dumps
- WorkingFuser/_diag/ — fuser launch failure diagnostics

11. Glossary

- **Photogrammetry:** Turning overlapping photos into a 3D model by solving camera positions and reconstructing geometry and textures.
- **Mesh / 3DML:** A 3D terrain artifact produced by PhotoMesh; serves as an intermediate for Reality Mesh and VBS4 export.
- **Fuser:** A worker process/computer that contributes CPU resources to accelerate photogrammetry.
- **UNC Path:** A network path in the form \\host\share\folder.
- **COTS:** Commercial off-the-shelf software (e.g., PhotoMesh).

12. Reference

Source material: 'OneClick Brief 3SEP V4' slide deck and 'OneClick_Fuser_Technical_Guide.md' internal technical reference.